Logarithms and Their Properties

ANSWER KEY

Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32 \;= 5$ **b** $\log_3 \frac{1}{9} \;= -2$ **c** $\log 1000 \;= 3$ **d** $\ln e^7 \;= 7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

$\log_5 18 = \frac{\ln 18}{\ln 5} \approx \frac{2.8904}{1.6094} \approx 1.796$.

3 State the domain of $f(x) = \log(x-4)$ and the inverse function of $g(x) = 2^x$.

Domain of f: $x > 4$. Inverse of g: $g^{-1}(x) = \log_2 x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

$3\ln x + \frac{1}{2}\ln y - 2\ln z$.

5 Write as a single logarithm: $2\log x + 3\log y - \log z$.

$\log\left(\frac{x^2y^3}{z}\right)$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6 \;= 1.114$ **b** $\log_b 8 \;= 1.293$ **c** $\log_b 1.5 \;= 0.252$

Graphs of logarithmic functions

7 For $y = \log_2(x+4) - 1$, state:

a the domain $x > -4$

b the vertical asymptote $x = -4$

c the x-intercept $0 = \log_2(x+4) - 1 \Rightarrow x = -2$

8 Describe how the graph of $g(x) = -\ln(x-2) + 3$ is obtained from $y = \ln x$.

Shift right 2 units, reflect across the x-axis, then shift up 3 units.

Logarithmic scales and applications

9 The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

a An earthquake has $M = 6$. Express I in terms of I_0. $I = I_0 \cdot 10^6$.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake? Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.

10 Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?
$\Delta\beta = 10\log(2) \approx 10(0.3010) \approx 3.01$ dB.

More practice with logarithm properties

11 Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.
$\log_2 8 + 2\log_2 x - \frac{1}{3}\log_2 y = 3 + 2\log_2 x - \frac{1}{3}\log_2 y$.

12 Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.
$\ln(16^{1/2}) + \ln 3 - \ln 2 = \ln 4 + \ln 3 - \ln 2 = \ln\left(\frac{4 \cdot 3}{2}\right) = \ln 6$.

Logarithms as inverse functions

13 The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.
Since $y = \log_2 x$ is defined as the inverse of $y = 2^x$, swapping the roles of x and y (which is exactly what reflecting across the line $y = x$ does geometrically) transforms one graph into the other.

Solving simple logarithmic equations

14 Solve $\log_3(x + 2) = 4$ for x.
$x + 2 = 3^4 = 81 \Rightarrow x = 79$.

15 Solve $2\log x = \log(x + 6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.
$2\log x = \log(x + 6) \Rightarrow \log(x^2) = \log(x + 6) \Rightarrow x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$. $x = 3$ or $x = -2$. Since $\log x$ requires $x > 0$, $x = -2$ is extraneous. Solution: $x = 3$.

The pH scale (a logarithmic application)

16 pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.
$pH = -\log(3.2 \times 10^{-5}) \approx -(-4.4949) \approx 4.49$.

17 Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.

$[H^+]_1 = 10^{-3}$, $[H^+]_2 = 10^{-6}$. Ratio $= \frac{10^{-3}}{10^{-6}} = 10^3 = 1000$. The pH-3 solution has 1000 times the hydrogen-ion concentration and is far more acidic (lower pH means more acidic).

Common mistakes with logarithm properties

18 Explain why $\log(x+y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

Using $x = y = 10$: $\log(10+10) = \log 20 \approx 1.301$, but $\log 10 + \log 10 = 1 + 1 = 2$. Since $1.301 \neq 2$, the two expressions are not equal — the correct property is $\log x + \log y = \log(xy)$, not $\log(x+y)$.

19 Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.

$\log_2(4x) - \log_2(2x) = \log_2\left(\frac{4x}{2x}\right) = \log_2(2) = 1$. Check with $x = 5$:
$\log_2(20) - \log_2(10) \approx 4.32 - 3.32 = 1 \checkmark$.